Building a Home SOC Lab with Wazuh SIEM

Log Monitoring, Threat Detection, and Attack Simulation

Introduction

Modern organizations rely on **Security Operations Centers (SOC)** to monitor infrastructure, detect cyber threats, and respond to security incidents. To gain practical experience in SOC monitoring and threat detection, I built a Home SOC Lab using Wazuh, an open source security monitoring platform.

This blog walks through how I built a Home SOC Lab in a virtual environment with a minimal setup, replicating key SOC monitoring workflows while keeping the infrastructure simple and accessible.

This project demonstrates several SOC monitoring capabilities:

- ***SIEM Deployment*** *using Wazuh*
- ***Endpoint Monitoring*** *using Wazuh Agents*
- ***File Integrity Monitoring (FIM)*** *to detect file changes*
- ***Windows Security Log Monitoring***
- ***PowerShell Activity Monitoring***
- ***Firewall Log Monitoring & Network Scan Detection***
- ***Vulnerability Detection*** *using Wazuh's vulnerability module*
- ***Threat Hunting and Security Event Investigation***

The lab simulates how security teams monitor endpoints, collect logs, and detect suspicious behavior.

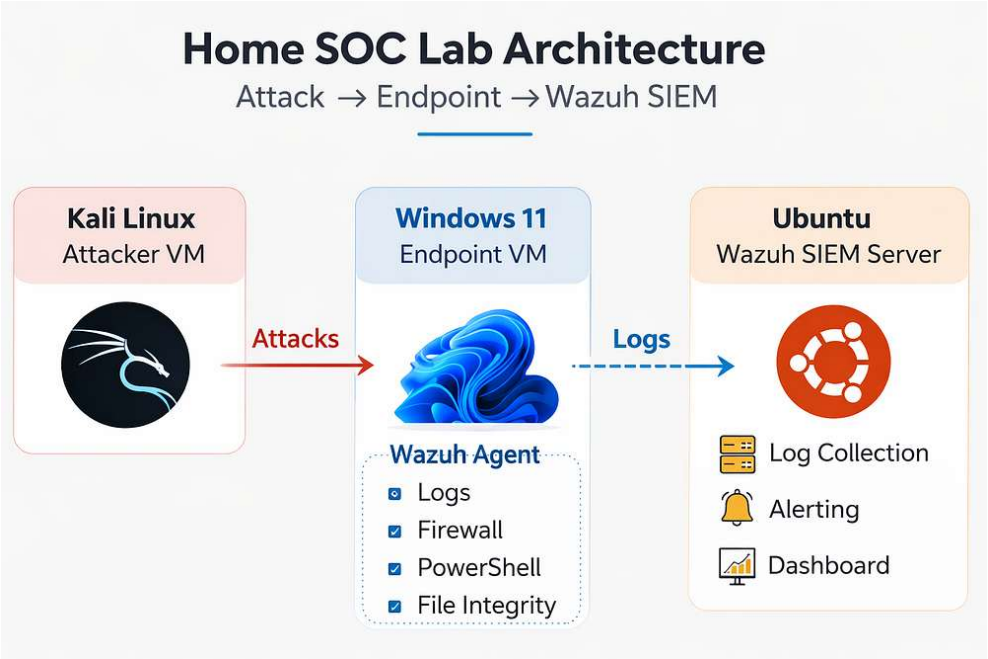
Lab Architecture

The entire lab environment was deployed using **VMware Workstation**, allowing multiple virtual machines to simulate an internal network environment, each performing a specific role.

Kali Linux: Attack simulation machine

Windows 11: Endpoint victim system

Ubuntu Server: Wazuh SIEM server



The attacker machine (Kali Linux) is used to simulate malicious activity against the Windows endpoint, while the Wazuh server on Ubuntu collects and analyzes logs generated by the endpoint.

Installing the Wazuh SIEM

Wazuh provides an automated installation script that installs all required components. It installs and configures the following components:

- Wazuh Manager
- Wazuh Indexer
- Wazuh Dashboard

```
Wazuh Installation Command: curl -sO https://packages.wazuh.com/
4.14/wazuh-install.sh && sudo bash ./wazuh-install.sh -a
```

(This script can be found on Wazuh.com)

Installation Issue Encountered

During the installation process, the setup repeatedly failed when installing the **Wazuh Dashboard**. The installation log displayed the following error:

ERROR: Wazuh dashboard installation failed

After this error occurred, the installer automatically removed previously installed components such as the Wazuh manager and indexer.

The problem was related to **disk allocation inside the Ubuntu virtual machine** Although VMware showed the disk size as **60 GB**, the Ubuntu operating system was only using a **19 GB root partition**.

Wazuh components require significant disk space, thus the installation process failed due to insufficient storage. To fix the issue, the root partition was expanded to use the full disk capacity.

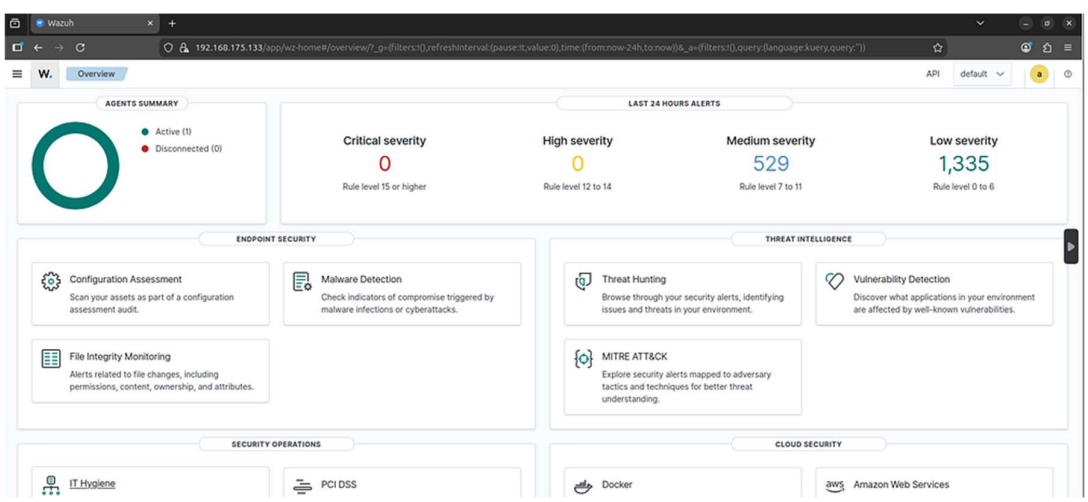
```
kushagra@kushagra: ~        kushagra@kushagra: ~        kushagra@kushagra: ~
kushagra@kushagra:~$ lsblk
NAME   MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
fd0      2:0    1     4K  0 disk
loop0    7:0    0     4K  1 loop /snap/bare/5
loop1    7:1    0    74M  1 loop /snap/core22/2292
loop2    7:2    0  18.5M  1 loop /snap/firmware-updater/210
loop3    7:3    0 251.7M  1 loop /snap/firefox/7766
loop4    7:4    0  91.7M  1 loop /snap/gtk-common-themes/1535
loop5    7:5    0  10.8M  1 loop /snap/snap-store/1270
loop6    7:6    0  48.1M  1 loop /snap/snapd/25935
loop7    7:7    0 531.4M  1 loop /snap/gnome-42-2204/247
loop8    7:8    0   576K  1 loop /snap/snapd-desktop-integration/343
sda      8:0    0    60G  0 disk
├─sda1   8:1    0   953M  0 part /boot/efi
└─sda2   8:2    0  19.1G  0 part /
sr0     11:0    1  99.5M  0 rom  /media/kushagra/CDROM
sr1     11:1    1   6.2G  0 rom  /media/kushagra/Ubuntu 24.04.4 LTS amd64
kushagra@kushagra:~$ sudo growpart /dev/sda 2
[sudo] password for kushagra:
CHANGED: partition=2 start=1953792 old: size=39987200 end=41940991 new: size=123875295 end=125829086
kushagra@kushagra:~$ sudo resize2fs /dev/sda2
resize2fs 1.47.0 (5-Feb-2023)
Filesystem at /dev/sda2 is mounted on /; on-line resizing required
old_desc_blocks = 3, new_desc_blocks = 8
The filesystem on /dev/sda2 is now 15484411 (4k) blocks long.

kushagra@kushagra:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs           790M  2.0M  788M   1% /run
/dev/sda2        58G   11G   45G  20% /
tmpfs           3.9G     0  3.9G   0% /dev/shm
tmpfs           5.0M  8.0K  5.0M   1% /run/lock
efivarfs        256K   63K  189K  25% /sys/firmware/efi/efivars
```

After resizing the filesystem, the Ubuntu system correctly detected the **full 60 GB disk space**, allowing the installation to proceed successfully.

The Wazuh web interface became accessible via: ***https://<server-ip>***

This dashboard acts as the **central monitoring console for the SOC lab**.

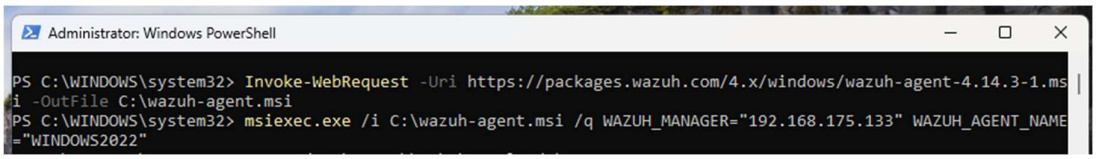


Wazuh Dashboard

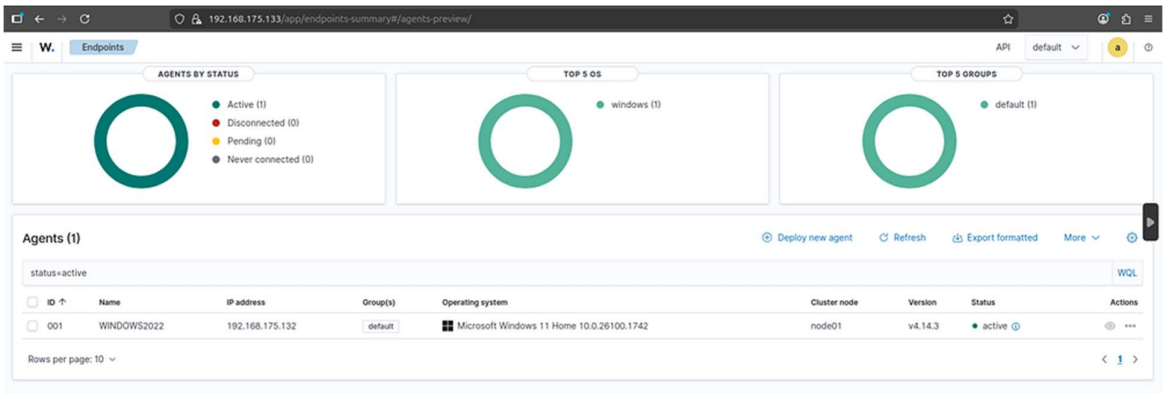
Installing the Wazuh Agent on Windows

To monitor the Windows endpoint, the **Wazuh agent** was installed. The agent collects logs from the endpoint and sends them to the Wazuh server.

Installation was performed using direct PowerShell commands.



Once installed, the Windows machine appeared as an **active agent in the Wazuh dashboard**.



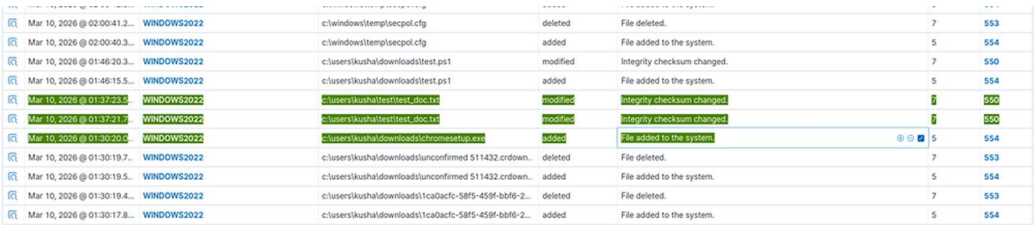
Active Agent

File Integrity Monitoring Setup

One of the key features implemented in this lab was **File Integrity Monitoring**. FIM detects file system activity such as:

- File creation
- File modification
- File deletion
- Permission changes

Specific directories such as the **Downloads folder and a custom test directory** were configured for monitoring. When files were modified or created in these directories, Wazuh generated alerts containing **File name, File path, Event type, Timestamp** etc.



FIM Logs

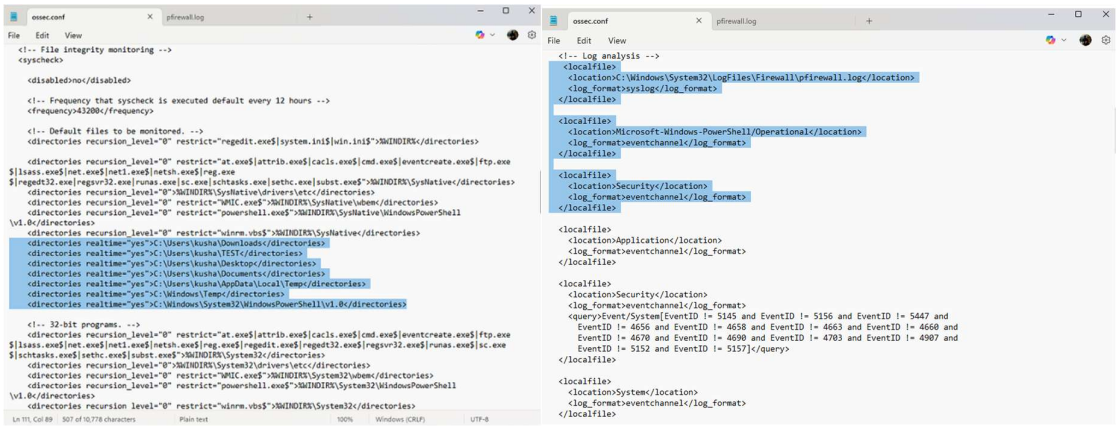
This demonstrated how SOC tools detect unauthorized file changes.

Monitoring Windows Logs

To improve visibility, the following Windows log sources were enabled:

- Windows Security logs
- System logs
- Application logs
- PowerShell operational logs
- Windows Firewall logs

These log sources were configured in the Wazuh agent configuration file ***(ossec.conf)*** using the ***<localfile>*** directive on the Windows endpoint.



Custom Rules

Simulating an Attack Using Nmap

To test detection capabilities, I simulated reconnaissance activity using **Nmap** from the Kali Linux machine.

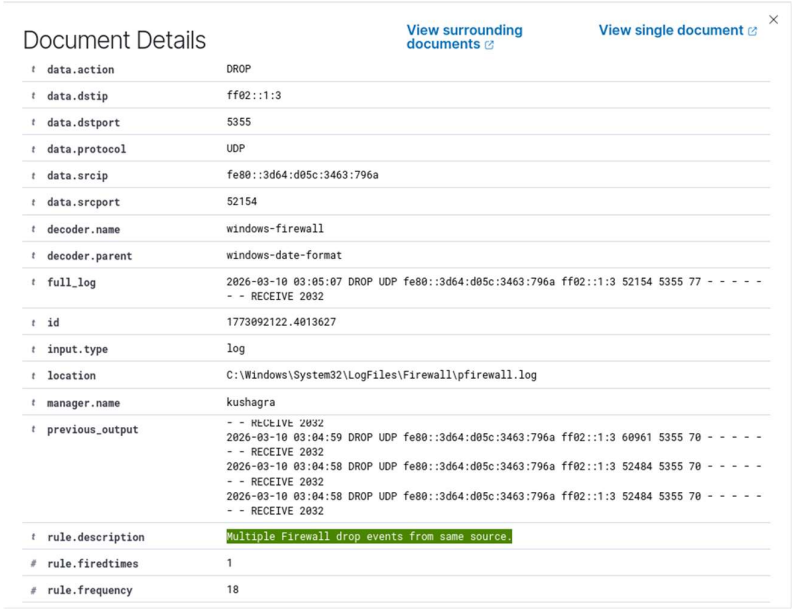
Example scan: *sudo nmap -sS -p 1–1000 <windows-ip>*

The scan attempted connections to multiple ports on the Windows system. Since the Windows firewall blocked these attempts, firewall log entries were generated and collected by the Wazuh agent.

Mar 10, 2026 @ 03:15:29.0...	WINDOWS2022	Multiple Firewall drop events from same source.
Mar 10, 2026 @ 03:05:22.1...	WINDOWS2022	Multiple Firewall drop events from same source.
Mar 10, 2026 @ 02:45:07.4...	WINDOWS2022	Multiple Firewall drop events from same source.
Mar 10, 2026 @ 02:35:07.9...	WINDOWS2022	Multiple Firewall drop events from same source.

Firewall Logs

Wazuh analyzed the firewall logs and triggered an alert rule **ID 4151.** This rule detects repeated connection attempts from a single source, a common indicator of **network scanning or reconnaissance activity**.



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data.action	DROP
data.dstip	ff02::1:3
data.dstport	5355
data.protocol	UDP
data.srcip	fe80::3d64:d05c:3463:796a
data.srcport	52154
decoder.name	windows-firewall
decoder.parent	windows-date-format
full_log	2026-03-10 03:05:07 DROP UDP fe80::3d64:d05c:3463:796a ff02::1:3 52154 5355 77 - - - - - - - RECEIVE 2032
id	1773092122.4013627
input.type	log
location	C:\Windows\System32\LogFiles\Firewall\pfirewall.log
manager.name	kushagra
previous_output	- - RECEIVE 2032 2026-03-10 03:04:59 DROP UDP fe80::3d64:d05c:3463:796a ff02::1:3 60961 5355 70 - - - - - - - RECEIVE 2032 2026-03-10 03:04:58 DROP UDP fe80::3d64:d05c:3463:796a ff02::1:3 52484 5355 70 - - - - - - - RECEIVE 2032 2026-03-10 03:04:58 DROP UDP fe80::3d64:d05c:3463:796a ff02::1:3 52484 5355 70 - - - - - - - RECEIVE 2032
rule.description	Multiple Firewall drop events from same source.
rule.firedtimes	1
rule.frequency	18

These alerts can be investigated in the **Threat Hunting section of the Wazuh dashboard**.

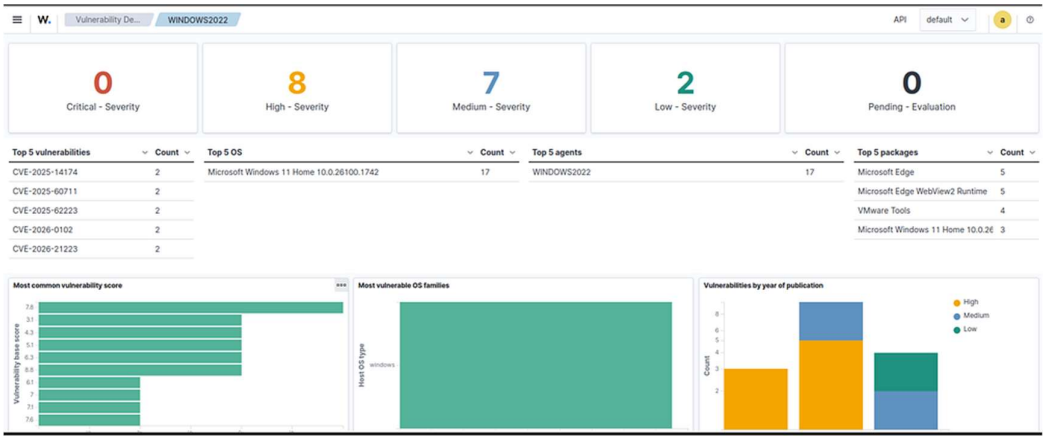
Vulnerability Detection

Beyond log monitoring, Wazuh provides additional security capabilities. The vulnerability detection module analyzes installed software and compares it with known vulnerabilities from databases such as the **National Vulnerability Database (NVD)**.

The dashboard highlights:

- Critical vulnerabilities
- High severity vulnerabilities
- Medium severity vulnerabilities
- Low severity vulnerabilities

It also identifies vulnerable software packages such as **Microsoft Edge, Windows components, and VMware Tools**.



Vulnerability Detection

Future Improvements

To further expand this lab, the following enhancements are planned:

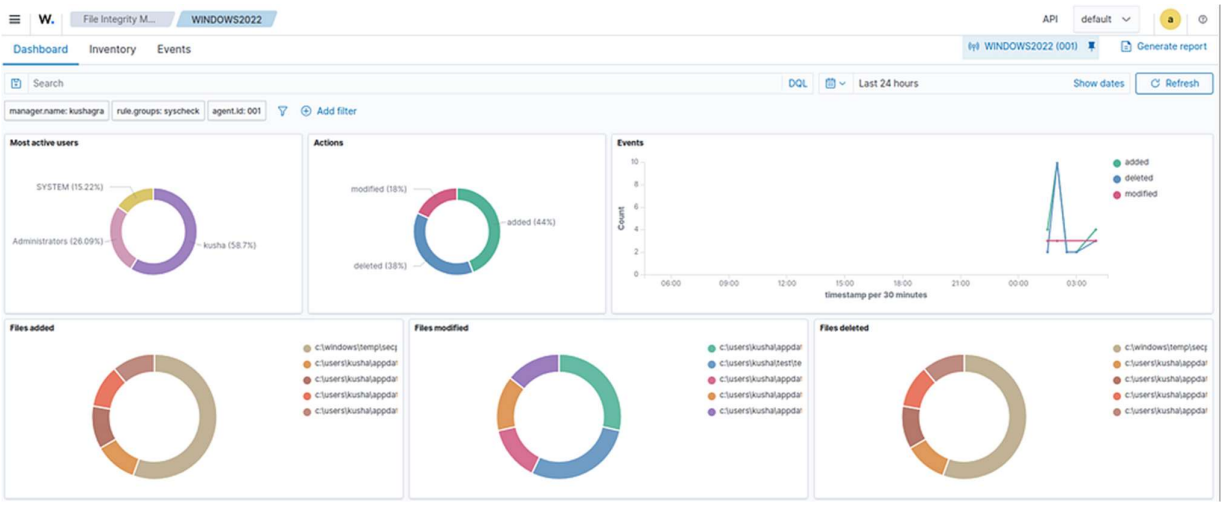
- Integrating **Sysmon for deeper endpoint telemetry**
- Creating **custom Wazuh detection rules**
- Simulating additional attack techniques
- Mapping alerts to the **MITRE ATT&CK framework**
- Testing detection of persistence and privilege escalation techniques

Conclusion

This lab demonstrated several important SOC concepts, including SIEM deployment, endpoint monitoring, security log analysis, attack simulation, and vulnerability detection. By building and troubleshooting this environment, I gained practical experience with log collection, threat detection, and security event investigation using a SIEM platform.

Developing a Home SOC Lab with Wazuh provides valuable insight into how modern security monitoring systems operate. By collecting logs from endpoints, analyzing them through a centralized SIEM, and simulating attack scenarios, it becomes easier to understand how SOC analysts identify, analyze, and respond to potential security threats.

Projects like this are an effective way to build practical cybersecurity skills while gaining experience with industry-relevant tools and real-world security workflows.



Reference tutorial used for guidance:
https://www.youtube.com/watch?v=Vm2_1_XEa5s

- Kushagra Gupta